

boundary. To identify such discontinuities, we employ the Pruned Exact Linear Time (PELT) changepoint detection algorithm [26], which efficiently detects multiple changepoints in time series data by minimizing a penalized cost function. PELT has been widely used for identifying structural breaks in sequential data [3] and is particularly well-suited for our task as it can detect abrupt shifts in mean performance without requiring prior specification of the number of changepoints. We apply PELT to the time-ordered sequence of faithfulness scores, allowing the algorithm to identify dates where model performance exhibits statistically significant drops.

It should be additionally noted that detecting the true pretraining cutoff is complicated by the fact that instruction-tuned models may refuse to answer questions beyond a declared cutoff date, despite having been pretrained on data from later time periods. To address this caveat, we also calculate average and cumulative refusal rates alongside faithfulness scores.

4 Results and Findings

We applied LLMagBench to a diverse set of LLMs to evaluate how well our methodology identifies training cutoffs across different scenarios. Our analysis reveals several distinct patterns in the relationship between model release dates, vendor- and mode-declared cutoffs, and empirically detected training boundaries. In the following subsections, we present selected cases that illustrate these patterns, demonstrating how LLMagBench can uncover information about training data freshness that may not be evident from public disclosures or direct model queries. Each case highlights a different combination of available information (release dates, declared cutoffs, self-reported dates) and our benchmark-derived estimates.

4.1 Grok-4: Single Cutoff Detection

Figure 1 illustrates the combined effect of this methodology when applied to xAI Grok-4. Blue dots represent faithfulness scores (0-2 scale, left y-axis) for individual questions ordered by event date, while red horizontal lines indicate mean faithfulness scores within segments identified by PELT. The green curve shows the cumulative average faithfulness score at each week. Red dashed vertical lines mark the detected changepoints (possible training boundaries), and the orange line (right y-axis) tracks cumulative refusals over time. The

model maintains high performance (faithfulness ≈ 1.7) from the beginning 2021 through late 2024, confirming comprehensive knowledge of events during this period. However, PELT identifies two closely localized changepoints in late 2024 and early 2025, where performance degrades progressively as refusal rates increase from 6.5% to 77.0%. This refusal pattern is characteristic of instruction-tuned models trained to decline questions about events beyond a declared cutoff date. The sharp rise in refusals coinciding with the drop in faithfulness scores suggests that Grok-4’s behavioral cutoff aligns closely with its actual pretraining data boundary. Given the overall performance characteristics, *the earliest probable cutoff date* for Grok-4 can be estimated at late 2024. As shown in Table 3, this estimate revealed by LLMagBench is significantly earlier than the model’s release date (August 2025) and it coincides with the xAI’s declarations of November 2024 as the cutoff for this model. Interestingly, this changepoint cannot be reliably determined by directly prompting the model, as in our tests Grok-4 self-reports a training cutoff of November 2023—more than a year earlier than our empirical estimate suggests.

4.2 Claude Sonnet 4: Multiple Partial Cutoffs

Figure 2 presents a different pattern for Anthropic’s Claude Sonnet 4. PELT identifies two distinct changepoints at February 2023 and December 2024, dividing the model’s performance into three segments with progressively declining mean faithfulness scores (1.25, 0.93, and 0.05, respectively) and increasing refusal rates (28.7%, 46.8%, and 95.1%).

The first changepoint in early 2023 is particularly notable, occurring well before the model provider’s declared cutoff of January 2025 and its release date of May 2025. This suggests that Claude Sonnet 4’s knowledge base was not uniformly updated throughout its training process. The extended intermediate segment (February 2023 to December 2024) exhibits moderate performance with nearly half of all questions refused, indicating partial knowledge coverage during this period. This pattern may indicate that the model in question underwent initial pretraining with a cutoff in early 2023, followed by selective knowledge updates or continued pretraining on limited data extending into late 2024.

The second changepoint at December 2024 aligns closely with the provider-declared cutoff of

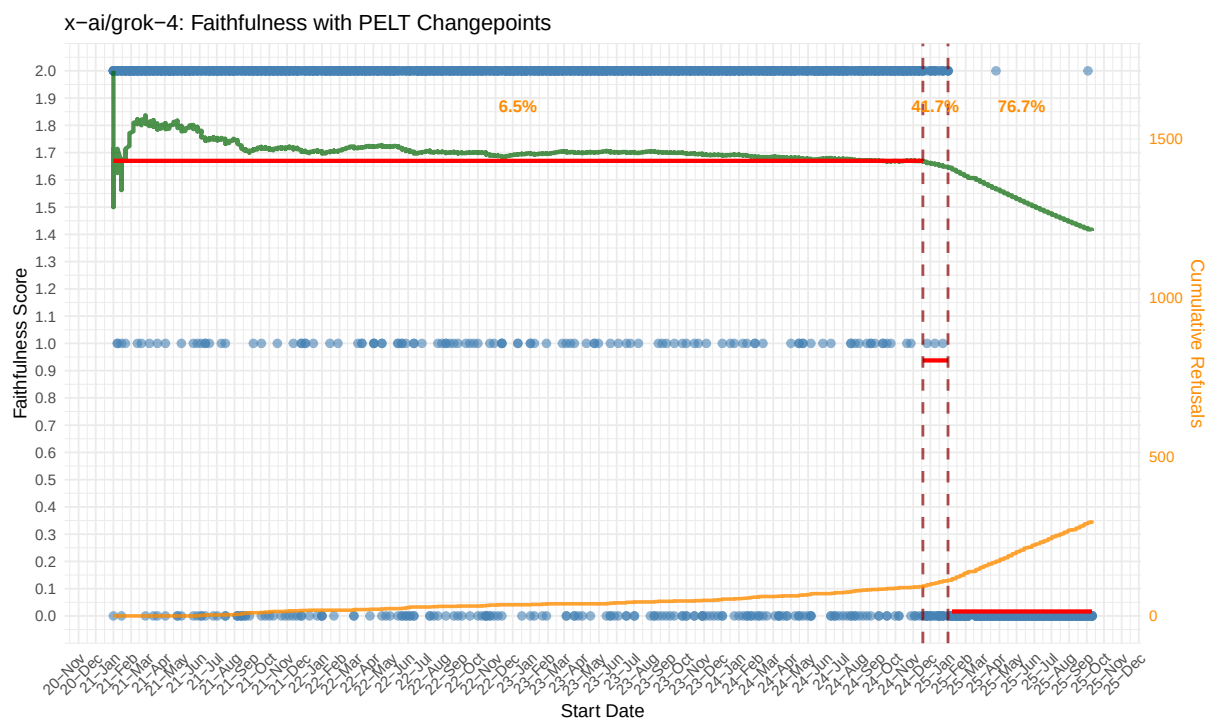


Figure 1: xAI Grok-4 training cutoff point estimation.

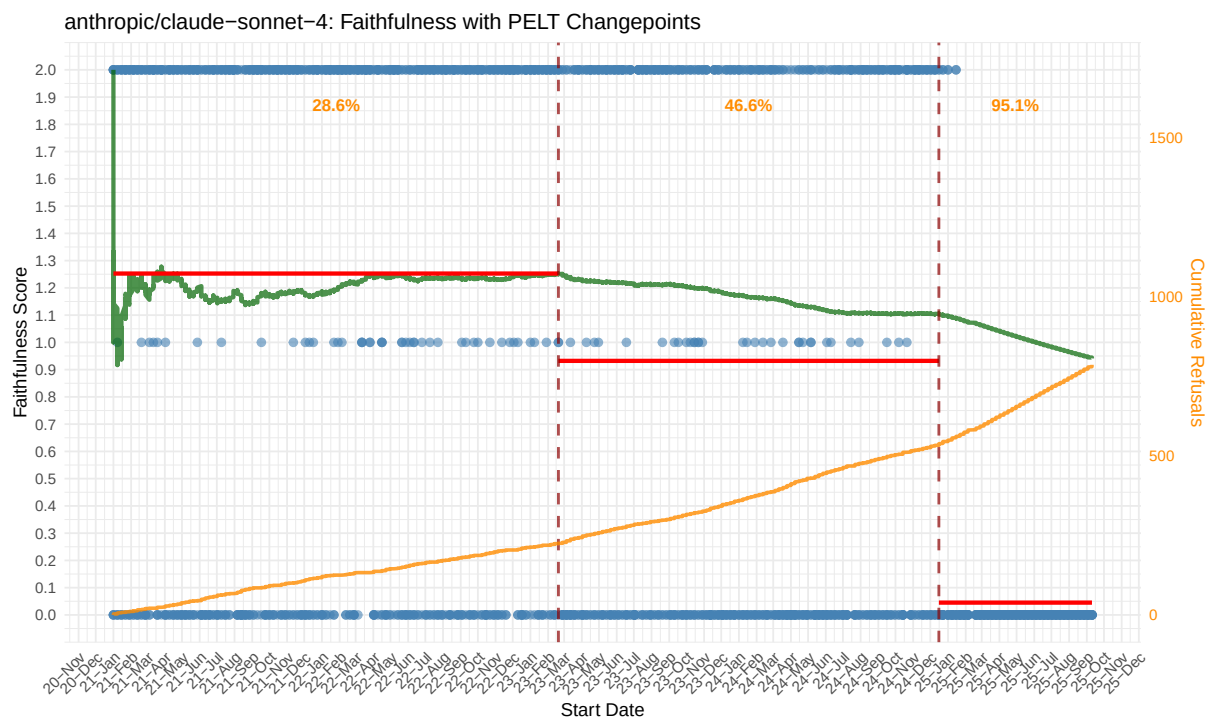


Figure 2: Claude Sonnet 4 training cutoff point estimation.

January 2025, where performance drops sharply and refusal rates exceed 95%. Unlike Grok-4, which maintained consistently high performance until its cutoff, Claude Sonnet 4 exhibits multiple partial knowledge boundaries possibly corresponding to different training phases. If this interpretation is correct, the declared January 2025 cutoff may obscure the significant knowledge degradation that began nearly two years earlier.

4.3 GPT-OSS-120B: Discrepancy Between Declared and Detected Cutoffs

Figure 3 illustrates yet another distinct pattern in training cutoff characteristics. PELT identifies a single changepoint at September 2023, dividing performance into two segments with mean faithfulness scores of 0.72 and 0.19, and refusal rates of 5.9% and 32.2%, respectively.

The benchmark-detected cutoff of September 2023 reveals a significant discrepancy with OpenAI’s declared cutoff of July 2024—nearly a full year later. This suggests that while the provider may have included some training data extending into mid-2024, the model’s reliable knowledge effectively ends in September 2023. The relatively modest initial faithfulness score of 0.72 (compared to Grok-4’s 1.7 or Claude Sonnet 4’s 1.25) indicates that even within its primary knowledge period, GPT-OSS-120B demonstrates lower overall performance on time-sensitive factual questions.

Notably, the model’s self-reported cutoff of September 2021 is dramatically earlier than both the provider’s declaration and our empirical estimate—approximately two years off from the benchmark-detected boundary. This extreme underestimation suggests either highly conservative instruction-tuning regarding knowledge claims, or a misalignment between the model’s self-assessment capabilities and its actual training data scope.

The post-cutoff segment shows a moderate increase in refusals (from 5.9% to 32.2%), but the refusal rate remains substantially lower than those observed in Grok-4 or Claude Sonnet 4. This indicates that GPT-OSS-120B continues to attempt answers for post-cutoff questions rather than reliably declining them, potentially leading to higher rates of hallucination about recent events. This case underscores the risk of relying on either provider declarations or model self-reports without empirical validation through benchmarks like LLMlag-Bench.

4.4 Gemma 3 Models: Model Size and Cutoff Detection Reliability

Figures 4 and 5 present results for Google’s Gemma 3 models in two different sizes (4B and 27B parameters), offering insights into how model capacity affects both overall performance and the reliability of cutoff detection.

The larger Gemma 3-27B model demonstrates a pattern similar to Claude Sonnet 4, with two changepoints at February 2023 and May 2024. The model exhibits stronger initial performance (faithfulness score of 1.11), followed by moderate performance in the intermediate segment (0.88), and finally sharp degradation after May 2024 (0.13 with 46.1% refusals). Notably, the benchmark-detected cutoff of May 2024 aligns closely with Google’s declared cutoff of August 2024—a discrepancy of only three months.

For the smaller Gemma 3-4B model PELT identifies a single changepoint at January 2024, with substantially lower faithfulness scores both before (0.36) and after (0.14) the cutoff. However, despite this low absolute performance, the changepoint remains clearly detectable, and refusal rates increase from near-zero (0.4%) to 12.6% after the cutoff. Notably, the post-cutoff refusal rate of 12.6% remains substantially lower than that of the larger 27B variant (46.1%), suggesting that smaller models may be more prone to hallucination about post-cutoff events. More generally, model size significantly impacts absolute performance on time-sensitive factual questions as the 4B model struggles even within its training data range.

Beyond identifying training cutoffs, LLMlag-Bench also provides an averaged overview of LLM performance on time-sensitive factual knowledge spanning 2021-2025. Table 2 summarizes the results across all evaluated models, ranked by average faithfulness score. The table presents several key metrics: overall average faithfulness (0-2 scale), overall refusal rate, the number and dates of detected changepoints, mean faithfulness in the first segment (representing performance within the model’s primary training period), and post-changepoint refusal rate (for the period after the last changepoint).

This ranking reveals substantial variation in model capabilities, with top-performing models like xAI’s Grok-4 (average faithfulness 1.41) and Grok-3 (1.37) demonstrating significantly stronger knowledge of recent events compared to lower-